

Amirhossain Arshadi Kalemeh

B.Sc. in Physics · Complex Systems · Network Science · Quantitative Biology



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PROFILE

Physics graduate of Isfahan University of Technology, ranked first in the department for four years, working where **network science**, **statistical physics** and **quantitative biology** meet. I test when network-complexity measures carry information beyond null models, build machine-learning surrogates for spectral properties of graphs, and model metabolic regulation in living systems, with research experience at the **Max Planck School Matter to Life** and **ICTP's Quantitative Life Sciences section**. I am seeking a PhD position in complex systems, network science or biophysics.

Complex networks Statistical physics Quantitative biology Nonlinear dynamics Graph machine learning Time-series analysis

19.50 / 20

Cumulative GPA, B.Sc. in Physics

1st

Department rank at IUT, four years running

20 / 20

In six core courses, incl. statistical mechanics

7th

National University Physics Olympiad, 2025

EDUCATION

B.Sc. in Physics Completed

Oct 2022 – 2026

Isfahan University of Technology (IUT), Iran · all coursework and examinations completed; degree certificate being issued

- GPA **19.50 / 20** · Ranked **1st** in the Department of Physics for four years · Full undergraduate scholarship
- Thesis: *Complexity of Real and Model Networks* (supervisor: Prof. Farhad Shahbazi)
- Coursework graded **20 / 20**: Statistical Mechanics, Advanced Statistical Mechanics, Critical Phenomena, Analytical Mechanics, Quantum Mechanics, Computational Physics

Diploma in Mathematics & Physics

Oct 2019 – Jun 2022

National Organization for Development of Exceptional Talents (NODET / Sampad), Isfahan

National Physics Olympiad Training

Jan 2020 – Sep 2021

Young Scholars Club, Tehran · IPhO team-selection training; top 80 nationwide

RESEARCH HIGHLIGHTS

COMPLEX NETWORKS

Complexity beyond the degree sequence

Against degree-preserving null models (10 enzyme networks, 20 centralities), **centrality entropy is largely explained by the degree sequence**; structural descriptors are not.

≈ null

 Centrality-entropy descriptors

≠ null

 Structural descriptors: σ_{corr} , clustering, modularity

B.Sc. thesis · manuscript in preparation [1]

COMPLEX NETWORKS

Small-world structure of enzyme networks

Low-rewiring Watts–Strogatz networks ($p = 0.01$) reproduce the centrality–correlation pattern of real enzyme networks better than Erdős–Rényi or Barabási–Albert models.

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Enzyme networks where WS matches best

9 / 10

Poster · ICTP Spring College in the Physics of Complex Systems 2026 [3]

GRAPH MACHINE LEARNING

Fast spectral-gap prediction with GNNs

A graph convolutional network on a k -nearest-neighbour “graph of graphs” predicts algebraic connectivity λ_2 from 11 cheap features, trained on 6,000 BA, ER and WS graphs.

6,000 graphs

3 random-graph models

11 input features

Accuracy on held-out graphs

≈ 95%

Manuscript in preparation [2]

QUANTITATIVE BIOLOGY

Metabolic feedback in *C. elegans*

Reversible glycolysis–gluconeogenesis model, informed by fluorescence data, linking FBP, ATP/ADP and NADH/NAD⁺ dynamics to energy balance across sleep–wake states.

Glucose

 $\xrightleftharpoons[\text{FBPase}]{\text{PFK}}$

FBP

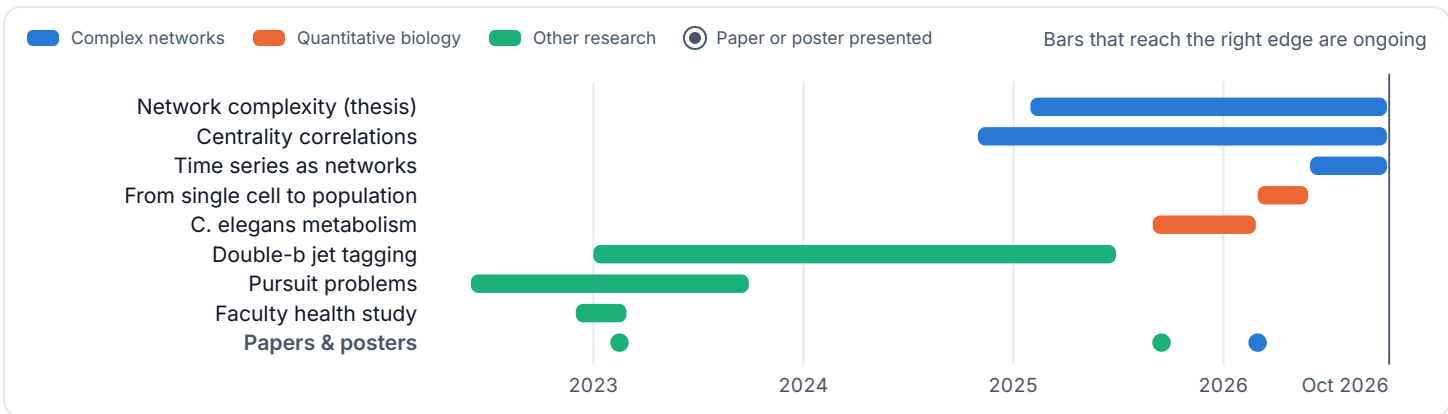
 $\xrightleftharpoons[\text{PCK}]{\text{PK}}$

Pyruvate

coupled to ATP/ADP and NADH/NAD⁺

MTL-URO · Max Planck School Matter to Life

RESEARCH TIMELINE



RESEARCH EXPERIENCE

COMPLEX NETWORKS & STATISTICAL PHYSICS

Complexity of Real and Model Networks B.Sc. thesis [1]

Feb 2025 – Present

Supervisor: Prof. Farhad Shahbazi · Isfahan University of Technology · with M. Tabatabaei

- Compared enzyme, metabolic and protein–protein interaction networks with ER, BA and WS models using Laplacian spectra and multi-centrality statistics: **low-rewiring small-world models fit enzyme and metabolic networks best**, while protein interaction networks are significantly more complex.
- Null-model test (10 ENZYMES networks, 20 centralities, configuration-model ensembles, Stouffer and Wilcoxon statistics): **centrality entropy is largely fixed by the degree sequence**, unlike structural descriptors (σ_{corr} , clustering, modularity); next: conditional entropy $S(c | k)$.

Correlations Between Centrality Measures Poster [3]

Nov 2024 – Present

Supervisor: Prof. Farhad Shahbazi · Isfahan University of Technology · with M. Tabatabaei

- Mapped correlations among 11 centralities across ER, BA and WS ensembles and proposed centrality entropy as a complexity indicator; enzyme networks match low-rewiring WS networks ($p = 0.01$) **best in 9 of 10 cases**.

Predicting Algebraic Connectivity with Graph Neural Networks Manuscript [2]

2026

With Yasin Jahangiri · Isfahan University of Technology

- A two-layer graph convolutional network on a k -nearest-neighbour “graph of graphs” (2,000 BA, ER and WS graphs each; 11 cheap features) predicts the Laplacian spectral gap λ_2 with **$\approx 95\%$ accuracy on held-out graphs**.

Time Series as Complex Networks

Jun 2026 – Present

Supervisor: Prof. Farhad Shahbazi · Isfahan University of Technology

- Comparing five time-series-to-network mappings (natural and horizontal visibility, ordinal-partition, quantile, proximity) in one ts2g2 pipeline, starting with AMZN daily closing prices.

QUANTITATIVE BIOLOGY

Modeling Metabolic Dynamics in the Sleep–Wake Cycle of *C. elegans*

Sep 2025 – Feb 2026

Max Planck School Matter to Life (MtL-URO), Dresden, Germany · Supervisors: Dr. Kathrin Laxhuber, Prof. Frank Jülicher

- Nonlinear model of glycolysis–gluconeogenesis (reversible glucose–FBP–pyruvate network via PFK, PK, FBPase, PCK) informed by fluorescence data; coupled FBP, ATP/ADP and NADH/NAD⁺ dynamics probe energy balance across sleep–wake states.

From Single Cell to Population ICTP internship

Mar – May 2026

Quantitative Life Sciences section, ICTP, Trieste, Italy · with Dr. Jacopo Grilli and Giuseppe Giuliani

- How bacterial cell physiology mediates growth trade-offs in fluctuating environments, through mathematical modelling and data analysis; re-derived a recent population-growth model, **identified an error in a key equation** and derived the corrected expression.

PARTICLE PHYSICS, DYNAMICS & OTHER RESEARCH

Double-b Jet Tagging

Jan 2023 – Jun 2025

Supervisor: Dr. Hamed Bakhshian, IUT

- ParticleNet (EdgeConv) on particle clouds to tag boosted double-b jets from $H \rightarrow aa \rightarrow b\bar{b}\mu^+\mu^-$, using jet substructure to improve signal–background separation.

Pursuit Problems in Two Modes

arXiv [4]

Jun 2022 – Sep 2023

With K. Bayati, R. Sharifian, M. M. Eftekhari

- Pursuit on lines and curves, generalized to N pairwise-chasing particles with a new relation; applied to moths circling a lamp.

Health Indicators of IUT Faculty

Paper [5]

Dec 2022 – Feb 2023

Supervisor: Dr. Mohammadreza Batavani, IUT

- Compared pre- and post-COVID-19 physiological health indicators of faculty members.

 PUBLICATIONS & PRESENTATIONS

- [1]

IN PREP.

A. Arshadi, M. Tabatabaei, F. Shahbazi. Entropy and Structural Interdependence in Network Complexity: A Null-Model Analysis.
- [2]

IN PREP.

A. Arshadi, Y. Jahangiri. Prediction of Laplacian Spectral Gap in Barabási–Albert, Erdős–Rényi and Watts–Strogatz Networks with Graph Neural Networks.
- [3]

POSTER

A. Arshadi, M. Tabatabaei, F. Shahbazi. Modeling the structure of enzyme networks using correlations between centrality measures. ICTP Spring College in the Physics of Complex Systems (SMR 4207), Trieste, Italy, 2026.
- [4]

PREPRINT · PAPER

A. Arshadi, K. Bayati, R. Sharifian, M. M. Eftekhari. Investigating the classical problem of pursuit, in two modes. arXiv:2309.02471 (2023); 9th Int. Conference on Physics, Mathematics and Development of Basic Science (2025).
- [5]

PAPER

M. R. Batavani, A. Arshadi, K. Bayati, A. M. Soltani. Investigating the physiological health indicators of faculty members of Isfahan University of Technology. 4th National & 1st International Conference on New Findings in Sport Sciences, Tehran (2023).

 HONORS & AWARDS

- 2025

Selected for Mtl-URO · Max Planck School Matter to Life; fully funded, about ten places a year
- 2025

7th place, National University Physics Olympiad · NODET
- 2022–26

Ranked 1st in the Department of Physics · four years, IUT
- 2023

1st place, 3rd IAP Research Tournament · IUST, Tehran
- 2022

Full undergraduate scholarship · IUT
- 2021

Top 80 nationwide, National Physics Olympiad · Young Scholars Club

 TEACHING & MENTORING

- 2024–now

Teaching Assistant, IUT · Complex Systems (graduate), Quantum Mechanics I & II, Mathematical Physics II, Physics II; problem sessions, computational exercises, mentoring
- 2022–25

Physics Olympiad Instructor · NODET (Sampad), Isfahan
- 2023–now

Student Guide for Incoming Students · IUT

 LEADERSHIP & SERVICE

- 2024–25

Chair, Physics Department Student Association · IUT
- 2024

Organizer, 4th IAP National Tournament · IUT and IUST
- 2024

Team Guide, Kazakhstan delegation · IPHo 2024, Isfahan
- 2022

Head of Physics Instructors · Isfahan Physics Olympiad Club

 SCHOOLS & CERTIFICATES

- 2026

ICTP Spring College in the Physics of Complex Systems · selected participant, Trieste, 16 Feb – 13 Mar; poster [3]
- Online

Supervised Machine Learning: Regression and Classification · Stanford (Coursera)
- Online

Python Data Structures · U. Michigan (Coursera) · Advanced Python Programming · Quera
- Online

Julia Programming Language: From Zero to Expert · Udemy · [all certificates](#) 

 SKILLS & LANGUAGES

PROGRAMMING

- Python

Advanced
- Machine learning

Intermediate
- Julia

Intermediate
- SQL (SQLite), CSS, JSON

Familiar

LANGUAGES

Persian: native · English: fluent

LIBRARIES & TOOLS

PyTorch, TensorFlow, scikit-learn, NetworkX, Matplotlib, QuTiP, ts2g2 · Mathematica, Maple · Linux, Git, LaTeX

THEORY & METHODS

Statistical mechanics and critical phenomena · network analysis (centralities, spectra, null models) · nonlinear dynamics and ODE models · graph neural networks · information theory

 REFERENCES

Prof. Farhad Shahbazi

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B.Sc. thesis supervisor

Dr. Hamed Bakhshian

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Jet-tagging research supervisor

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Academic records: transcript and grading scale available on request · [certificate repository](#)